CO2 ASSESSMENT TOOL -1

ANALYTICAL PROBLEM SOLVING

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QUESTION 1:

1. RSA Encryption Analysis .Given p=11, q=13

a. Compute n φ(n)

b. Choose e=7 and find the private key d

c. Encrypt the message M=9

d. Decrypt the ciphertext and verify the result

e. Analyze how changing e affects security

SOLUTION:

Given:

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

(a) Compute $\boldsymbol{n}$and $\boldsymbol{\phi}(\boldsymbol{n})$

$$n = p \times q = 11 \times 13 = 143$$
$$\phi(n) = (p-1)(q-1) = 10 \times 12 = 120$$

Answer:

- $n = 143$
- $\phi(n) = 120$

(b) Find the private key $\boldsymbol{d}$

- Need:

$$d \times e \equiv 1 \pmod{120}$$

Since

$$7 \times 103 = 721$$
$$721 \mod 120 = 1$$

Therefore,

$d = 103$

Private Key: $(103, 143)$

(c) Encrypt the message

Formula: $C = M^e \mod n$

$$C = 9^7 \mod 143$$

Using modular arithmetic, $9^7 \mod 143 = 48$
Ciphertext = 48

(d) Decrypt the ciphertext

Formula:

$$M = C^d \mod n$$
$$M = 48^{103} \mod 143 = 9$$

Recovered message = **9**

Hence,

Original Message = Decrypted Message

Verification successful.

(e) Effect of changing e

- Smaller e(like 3) → Faster encryption but may introduce vulnerabilities if used incorrectly.
- Moderate value (65537) → Most commonly used because it balances speed and security.
- Larger e→ Slower encryption with little additional practical security.

Conclusion:
Choosing $e = 65537$is recommended because it offers good efficiency and strong security.

QUESTION 2:

Block Cipher Mode Evaluation .Given plaintext blocks:

P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)

b. Compare the ciphertext outputs

c. Analyze which mode hides patterns better and explain why

SOLUTION:

Given

Plaintext blocks $P = [1010, 1010, 1111, 1010]$

Key $K = 1100$

IV 0000

Assume encryption is performed using XOR.

(a) ECB Mode

Formula $\boldsymbol{C = P \oplus K}$

Block 1	**Block 2**	**Block 3**	**Block 4**
1010 ⊕ 1100	1010 ⊕1100	1111⊕1100	1010⊕1100
=0110	=0110	=0011	=0110

ECB Ciphertext

$$[0110, 0110, 0011, 0110]$$

CBC Mode

Formula $C_i = (P_i \oplus C_{i-1}) \oplus K$

Initial IV = 0000

Block 1	**Block 2**	**Block 3**	**Block 4**
1010⊕0000=1010	1010⊕0110=1100	1111⊕0000=1111	1010⊕0011=1001
1010⊕1100=0110	1100⊕1100=0000	1111⊕1100=0011	1001⊕1100=0101

C1=0110 C2=0000 C3=0011 C4=0101

CBC Ciphertext :[0110, 0000, 0011, 0101]

(b) Comparison

ECB	CBC
0110	0110
0110	0000
0011	0011
0110	0101

Repeated plaintext blocks produce repeated ciphertext only in ECB.

(c) Which hides patterns better?

CBC Mode

Reason:

- Every block depends on the previous ciphertext block.
- Identical plaintext blocks generate different ciphertext.
- Better protection against pattern analysis.

QUESTION3 :

3. DES vs 3-DES vs Blowfish Performance Study

Given:

a. DES key size = 56 bits, time = 2 ms/block

b. 3-DES key size = 168 bits, time = 6 ms/block

c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm

ii.Analyze trade-offs between security and performance

iii.Recommend the best algorithm for secure real-time communication

SOLUTION:

Given

Algorithm	**Time/block**
DES	**2 ms**
3-DES	**6 ms**
Blowfish	**1 ms**

1000 blocks

(i) Total Encryption Time

DES

$$1000 \times 2 = 2000ms = 2s$$

3-DES

$$1000 \times 6 = 6000ms = 6s$$

Blowfish

$$1000 \times 1 = 1000ms = 1s$$

Algorithm	Total Time
DES	2 s
3-DES	6 s
Blowfish	1 s

(ii) Security vs Performance

Algorithm	**Security**	**Performance**
DES	Low (56-bit key)	Fast
3-DES	Very High	Slow
Blowfish	High	Very Fast

(iii) Recommendation

Blowfish

Reason:

- High security ,Fastest encryption ,Suitable for secure real-time communication

QUESTION 4:

Diffie-Hellman Key Exchange Analysis

Given:

a. p=23, g=5

b. User A private key = 6

c. User B private key = 15

d. Compute public keys for A and B

e. Compute the shared secret key

f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange

g. Suggest a secure method to prevent the attack

SOLUTION:

Given

- $p = 23$
- $g = 5$
- A private key =6
- B private key =15

(d) Public Keys

User A

$$A = 5^6 \mod 23$$
$$5^6 = 15625$$
$$15625 \mod 23 = 8$$

Public key of A = 8

User B

$$B = 5^{15} \mod 23$$
$$= 19$$

Public key of B = 19

(e) Shared Secret

User A

$$K = 19^6 \bmod 23 = 2$$

User B

$$K = 8^{15} \bmod 23 = 2$$

Shared Secret Key = 2

(f) Man-in-the-Middle Attack

Attacker intercepts both public keys.

- A sends 8 → attacker replaces it with 10.
- B sends 19 → attacker replaces it with 7.

Now:

- A shares a secret with attacker.
- B shares another secret with attacker.

The attacker decrypts, modifies, and re-encrypts all messages without either party realizing it.

(g) Prevention

Use:

- Digital Signatures
- Digital Certificates (PKI)
- Authenticated Diffie-Hellman
- TLS/SSL

These authenticate the communicating parties and prevent Man-in-the-Middle attacks.

<u>QUESTION 5:</u>

5. ECC vs RSA Comparison

Given:

a. RSA key size = 1024 bits

b. ECC key size = 160 bits (equivalent security)

c. Device processing capacity = limited (IoT device)

d. Analyze computational cost for both systems

e. Estimate which consumes less power and memory

f. Justify which cryptosystem is better for the given scenario with reasoning

<u>SOLUTION:</u>

Given

- RSA key = 1024 bits
- ECC key = 160 bits
- IoT device with limited processing capacity

(d) Computational Cost

RSA

- Large key size
- Slower computations
- Higher CPU usage

ECC

- Smaller key
- Faster calculations
- Lower computational cost

(e) Power and Memory Consumption

Feature	RSA	ECC
Key Size	1024 bits	160 bits
Memory	High	Low
Power Usage	High	Low
Speed	Slower	Faster

ECC requires less memory and consumes less power than RSA.

(f) Best Cryptosystem

ECC (Elliptic Curve Cryptography)

Reasons:

- Provides security equivalent to RSA with much smaller keys.
- Faster execution on resource-constrained devices.
- Lower memory and power consumption.
- Ideal for IoT devices and embedded systems.

Conclusion: ECC is the preferred cryptosystem for the given IoT scenario because it offers strong security while minimizing computational overhead, energy usage, and memory requirements.